NICOLÁS SCHKURKO

Software Developer — Backend, Integrations & Applied AI

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PROFESSIONAL PROFILE

Software developer with 4 years of experience building backend services, integrations and automation for marketing agencies. Fourth-year Information Systems Engineering student at UTN FRBA, with a background as an Electronics Technician. Recent work focuses on LLM agents, RAG pipelines, and open-source tooling for AI-assisted development. I document what I build so someone else can maintain it.

WORK EXPERIENCE

Nucleo Marketing Oct. 2022 – Present · Remote

AI & Software Developer (primary role)

- Deployed 3 Zoho CRM platforms end to end for different clients, plus channel unification services (Kommo, Wati) across multiple accounts.
- Built 2 conversational bots in n8n over the OpenAI API, handling up to 70 queries per day.
- Built RAG pipelines over pgvector (Supabase) for multi-format document intake and semantic search, with answers grounded in the ingested corpus.
- Designed relational schemas (PostgreSQL, MySQL, SQL Server) and analytical queries on BigQuery; built 7 dashboards (5 Power BI, 2 Looker Studio).
- Integrated third-party REST APIs and bidirectional webhooks; wrote middleware and automation in Node.js and JavaScript.

Coordline (Precision Marketing) Jan. 2023 – Present · Remote

Software Developer & Process Analyst (contractor)

- Built a WhatsApp Business API bot for direct support, lead qualification and automatic logging into Microsoft Dynamics CRM.
- Configured contact automation, audience segmentation and lifecycle rules in Dynamics CRM.

SELECTED PROJECTS

Spec-Growth (2026, MIT): spec-anchored change management for AI coding agents

- Method for developing with agents without losing control: every behavior change travels through a spec, a deterministic gate and earned evidence, and the agent writes a technical brief the developer must explain before a change closes.
- Measured with a frozen conformance suite: 17% fewer tokens per batch, up to 60% fewer on the heaviest task.

Foco (2026): local-first Android screen-time app

- Kotlin and Jetpack Compose, no accounts and no servers: per-app usage tracking, time limits, and an optional friction overlay.

LGR100-SteamVR (2026): SteamVR driver for an abandoned headset

- Reverse-engineered the HID protocol from a firmware dump and verified it against hardware; 3DOF tracking and stereoscopic rendering on Windows.

Distributed Microservices Platform (2026): annual project, UTN FRBA

- Co-developed 4 Spring Boot microservices with one MySQL database per service, RabbitMQ messaging, Docker Compose and GitHub Actions CI.
- Owned the incentives domain (profiles, ranking, bulk ingestion) and the repository's technical backlog documents.

TECHNICAL SKILLS

Languages: Java, Go, JavaScript/TypeScript, C/C++, SQL, Kotlin

Backend & Data: Spring Boot, Node.js, REST APIs, PostgreSQL, MySQL, SQL Server, BigQuery, pgvector

Cloud & DevOps: Google Cloud, Cloudflare, Docker, GitHub Actions, RabbitMQ, OAuth 2.0, Git

Applied AI: OpenAI API, Anthropic API, n8n, RAG, agent design, Vertex AI Vision (helmet detection POC)

Systems & Embedded: ATmega328P, ESP8266, PlatformIO, KiCad, UART, SPI, I2C, HID

EDUCATION

UTN FRBA Mar. 2023 – Dec. 2029 (expected)

Information Systems Engineering — 4th year

Instituto San José (A-355) 2017 – 2022

Electronics Technician

Languages: Spanish (native) · English (B2) · Portuguese (basic)